

Switch

A control statement that matches one variable against multiple fixed values, executing the matching block - a cleaner alternative to long else if chains.

When to use:

Use Switch case when you're checking one variable against many possible fixed values (like a day number, a menu choice, a grade letter) - especially when there are more than 2-3 alternatives, making

else if chains messy to read.

How to use:

Traditional

```
int day = 3;
switch(day) {
    case 1:
        S.o.p("Mon");
        break;
    case 2:
        S.o.p("Tue");
        break;
    case 3:
        S.o.p("Wed");
        break;
    default:
        S.o.p("Invalid");
}
```

Modern form

```
String dayName = switch(day) {
    case 1 → "Mon";
    case 2 → "Tue";
    case 3 → "Wed";
    default → "Invalid";
};
```

* Switch works with int, char, String, enum (not double/~~boolean~~)

Where to use:

Anywhere you're mapping a single value to a fixed set of outcomes - menu systems, day/month names, grade brackets, simple calculations (+, -, *, / as cases).

Why:

* Without break, execution falls through into the next case(s) - intentional, allows grouping cases with shared behaviour.

* New arrow syntax (→) doesn't fall through by default - removes that bug risk.

* double not allowed - floating point imprecision makes exact matching unreliable.

* boolean not allowed - only 2 values, if-else already handles it better.

* Switch can be faster than else if chains - compiler can optimize into a jump table.

Real-life Examples:

Scenario

ATM menu options

Calculator operator(+, -, *, /)

Day-of-week lookup

Grade letter $\rightarrow$ GPA

Switch use

Switch on choice number

Switch on char

Switch on day number

Switch on char.